

HALF ADDER:

CHIP HalfAdder {

 IN a, b; // 1-bit inputs

 OUT sum, // Right bit of a + b

 carry; // Left bit of a + b

PARTS:

Xor(a=a, b=b, out=sum);

And(a=a, b=b, out=carry);

}

NAND2Tetris / Hardware Simulator / HalfAdder.hdl

HDL ☐ --LBSt Project 2 HalfAdder Chip HalfAdder Eval Reset Clock: 0

```
1 CHIP HalfAdder {
2   IN a, b; // 1-bit inputs
3   OUT sum, // Right bit of a + b
4   carry; // Left bit of a + b
5
6   PARTS:
7     Xor(a=a, b=b, out=sum);
8     And(a=a, b=b, out=carry);
9
10 }
```

Input pins

a 0

b 0

Output pins

sum 0

carry 0

Test Steps: 0 Outputs: 0 Slow Fast

NAND2Tetris / Hardware Simulator / HalfAdder.hdl

HDL ☐ --LBSt Project 2 HalfAdder Chip HalfAdder Eval Reset Clock: 0

```
1 CHIP HalfAdder {
2   IN a, b; // 1-bit inputs
3   OUT sum, // Right bit of a + b
4   carry; // Left bit of a + b
5
6   PARTS:
7     Xor(a=a, b=b, out=sum);
8     And(a=a, b=b, out=carry);
9
10 }
```

Input pins

a 1

b 0

Output pins

sum 1

carry 0

Test Steps: 0 Outputs: 0 Slow Fast

NAND2Tetris / Hardware Simulator / HalfAdder.hdl

HDL ☐ --LBSt Project 2 HalfAdder

Chip HalfAdder Eval Reset Clock: 0

```

1 CHIP HalfAdder {
2   IN a, b;    // 1-bit inputs
3   OUT sum,    // Right bit of a + b
4     carry;    // Left bit of a + b
5
6   PARTS:
7     Xor(a=a, b=b, out=sum);
8     And(a=a, b=b, out=carry);
9
10 }

```

Input pins
a 0
b 1
Output pins
sum 1
carry 0

Test Steps: 0 Outputs: 0 Slow Fast

NAND2Tetris / Hardware Simulator / HalfAdder.hdl

HDL ☐ --LBSt Project 2 HalfAdder

Chip HalfAdder Eval Reset Clock: 0

```

1 CHIP HalfAdder {
2   IN a, b;    // 1-bit inputs
3   OUT sum,    // Right bit of a + b
4     carry;    // Left bit of a + b
5
6   PARTS:
7     Xor(a=a, b=b, out=sum);
8     And(a=a, b=b, out=carry);
9
10 }

```

Input pins
a 1
b 1
Output pins
sum 0
carry 1

Test Steps: 0 Outputs: 0 Slow Fast

FULL ADDER:

CHIP FullAdder{

IN a, b, c; // 1-bit inputs

OUT sum, // Right bit of a + b + c

carry; // Left bit of a + b + c

PARTS:

Xor(a=a , b=c , out=ab);

Xor(a=ab , b=c , out=sum);

And(a=a , b=b , out=c1);

And(a=a , b=c , out=c2);

And(a=b , b=c , out=c3);

Or(a=c1 , b=c2 , out=c12);

Or(a=c12 , b=c3 , out=carry);

}

NAND2Tetris / Hardware Simulator / FullAdder.hdl

Test Steps: 0 Outputs: 0 Slow Fast

Chip FullAdder Eval Reset Clock: 0

Input pins

a 0

b 0

c 0

Output pins

sum 0

carry 0

Internal pins

ab 0

c1 0

```
1 CHIP FullAdder {
2   IN a, b, c; // 1-bit inputs
3   OUT sum,    // Right bit of a + b + c
4       carry;  // Left bit of a + b + c
5
6   PARTS:
7     Xor(a=a, b=c, out=ab);
8     Xor(a=ab, b=c, out=sum);
9     And(a=a, b=b, out=c1);
10    And(a=a, b=c, out=c2);
11    And(a=b, b=c, out=c3);
12    Or(a=c1, b=c2, out=c12);
13    Or(a=c12, b=c3, out=carry);
14 }
15 }
```

NAND2Tetris / Hardware Simulator / FullAdder.hdl

Test Steps: 0 Outputs: 0 Slow Fast

Chip FullAdder Eval Reset Clock: 0

Input pins

a 1

b 1

c 1

Output pins

sum 1

carry 1

Internal pins

ab 0

c1 0

```
1 CHIP FullAdder {
2   IN a, b, c; // 1-bit inputs
3   OUT sum,    // Right bit of a + b + c
4       carry;  // Left bit of a + b + c
5
6   PARTS:
7     Xor(a=a, b=c, out=ab);
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15 }
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NAND2Tetris / Hardware Simulator / FullAdder.hdl

Test Steps: 0 Outputs: 0 Slow Fast

Chip FullAdder Eval Reset Clock: 0

Input pins

a 0

b 1

c 1

Output pins

sum 0

carry 1

Internal pins

ab 1

c1 0

```
1 CHIP FullAdder {
2   IN a, b, c; // 1-bit inputs
3   OUT sum,    // Right bit of a + b + c
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7     Xor(a=a, b=c, out=ab);
8     Xor(a=ab, b=c, out=sum);
9     And(a=a, b=b, out=c1);
10    And(a=a, b=c, out=c2);
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NAND2Tetris / Hardware Simulator / FullAdder.hdl

HDL ☐ --LBSt Project 2 FullAdder Chip FullAdder Eval Reset Clock: 0

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15 }
```

Input pins
a 0
b 1
c 0

Output pins
sum 0
carry 0

Internal pins
ab 0
c1 0

Test Steps: 0 Outputs: 0 Slow Fast

NAND2Tetris / Hardware Simulator / FullAdder.hdl

HDL ☐ --LBSt Project 2 FullAdder Chip FullAdder Eval Reset Clock: 0

```
1 CHIP FullAdder {
2   IN a, b, c; // 1-bit inputs
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11    And(a=b, b=c, out=c3);
12    Or(a=c1, b=c2, out=c12);
13    Or(a=c12, b=c3, out=carry);
14
15 }
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Input pins
a 0
b 0
c 1

Output pins
sum 0
carry 0

Internal pins
ab 1
c1 0

Test Steps: 0 Outputs: 0 Slow Fast

NAND2Tetris / Hardware Simulator / FullAdder.hdl

HDL ☐ --LBSt Project 2 FullAdder Chip FullAdder Eval Reset Clock: 0

```
1 CHIP FullAdder {
2   IN a, b, c; // 1-bit inputs
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11    And(a=b, b=c, out=c3);
12    Or(a=c1, b=c2, out=c12);
13    Or(a=c12, b=c3, out=carry);
14
15 }
```

Input pins
a 1
b 0
c 0

Output pins
sum 1
carry 0

Internal pins
ab 1
c1 0

Test Steps: 0 Outputs: 0 Slow Fast

HDL

--LBSt

Project 2

FullAdder

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Chip FullAdder

Eval Reset Clock: 0

```

1 CHIP FullAdder {
2   IN a, b, c; // 1-bit inputs
3   OUT sum,    // Right bit of a + b + c
4     carry;    // Left bit of a + b + c
5
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7     Xor(a=a, b=b, out=ab);
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9     And(a=a, b=b, out=c1);
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11    And(a=b, b=c, out=c3);
12    Or(a=c1, b=c2, out=c12);
13    Or(a=c12, b=c3, out=carry);
14
15 }

```

Input pins

a 1

b 1

c 0

Output pins

sum 1

carry 1

Internal pins

ab 1

c1 1

Test

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Steps: 0 Outputs: 0

Slow Fast

HDL

--LBSt

Project 2

FullAdder

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Chip FullAdder

Eval Reset Clock: 0

```

1 CHIP FullAdder {
2   IN a, b, c; // 1-bit inputs
3   OUT sum,    // Right bit of a + b + c
4     carry;    // Left bit of a + b + c
5
6   PARTS:
7     Xor(a=a, b=b, out=ab);
8     Xor(a=ab, b=c, out=sum);
9     And(a=a, b=b, out=c1);
10    And(a=a, b=c, out=c2);
11    And(a=b, b=c, out=c3);
12    Or(a=c1, b=c2, out=c12);
13    Or(a=c12, b=c3, out=carry);
14
15 }

```

Input pins

a 1

b 0

c 1

Output pins

sum 1

carry 1

Internal pins

ab 0

c1 0

Test

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Steps: 0 Outputs: 0

Slow Fast